

Used Car Price Prediction Using Machine Learning

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Data Mining

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1. Problem Statement and Background

The goal of this project is to predict used car prices using machine learning techniques. Used car pricing is a complex problem because multiple factors influence the value of a vehicle, including its age, engine size, mileage, and ownership history. Because of this, pricing can vary significantly even between similar vehicles.

Accurately predicting used car prices is important for both buyers and sellers. Buyers want to avoid overpaying, while sellers want to price their vehicles competitively. Traditional pricing methods rely on manual estimation or limited comparisons, which can be inconsistent. A data-driven approach using machine learning can provide more reliable and objective price estimates.

This problem is well suited for data mining because it involves identifying patterns in historical data and using those patterns to make predictions. In this project, the problem is treated as a regression task, where the goal is to predict a continuous value (price) based on multiple input features.

The scope of this project is limited to the dataset used, meaning the results reflect patterns found in that dataset and may not fully represent all real-world car markets.

2. Dataset and Data Collection

The dataset used in this project contains approximately 10,000 records and includes multiple features describing used vehicles. The dataset was obtained from an online public data source (Kaggle).

Each record represents a used car and includes attributes that influence pricing. The target variable is **price_usd**, which represents the selling price of the vehicle.

Key features in the dataset include:

- Make year
- Mileage (km per liter, fuel efficiency)
- Engine size (cc)
- Fuel type
- Owner count
- Brand
- Transmission
- Color
- Service history
- Accidents reported
- Insurance validity

Engine size represents the power and performance of the vehicle, while mileage reflects fuel efficiency. Owner count provides information about how many previous owners the car had, which can affect perceived value. These features collectively help describe the overall condition and desirability of the vehicle.

3. Data Preprocessing

Before building machine learning models, the dataset was cleaned and prepared through several preprocessing steps.

First, the dataset was loaded into a DataFrame and inspected using basic functions such as `.head()`, `.info()`, and `.describe()`. This allowed for an understanding of the data structure and identification of any issues.

Next, categorical variables such as fuel type, brand, and transmission were converted into numerical values using encoding techniques. This step is necessary because machine learning models require numerical input.

The dataset was then divided into two parts:

- Features (X), which include all input variables
- Target variable (y), which is the price

After that, the dataset was split into training and testing sets using an 80/20 ratio. This step is important because it allows the model to be trained on one portion of the data and evaluated on unseen data, reducing the risk of overfitting.

```
[89]: df = pd.read_csv('used_car_price_dataset_extended.csv')
df.head()
```

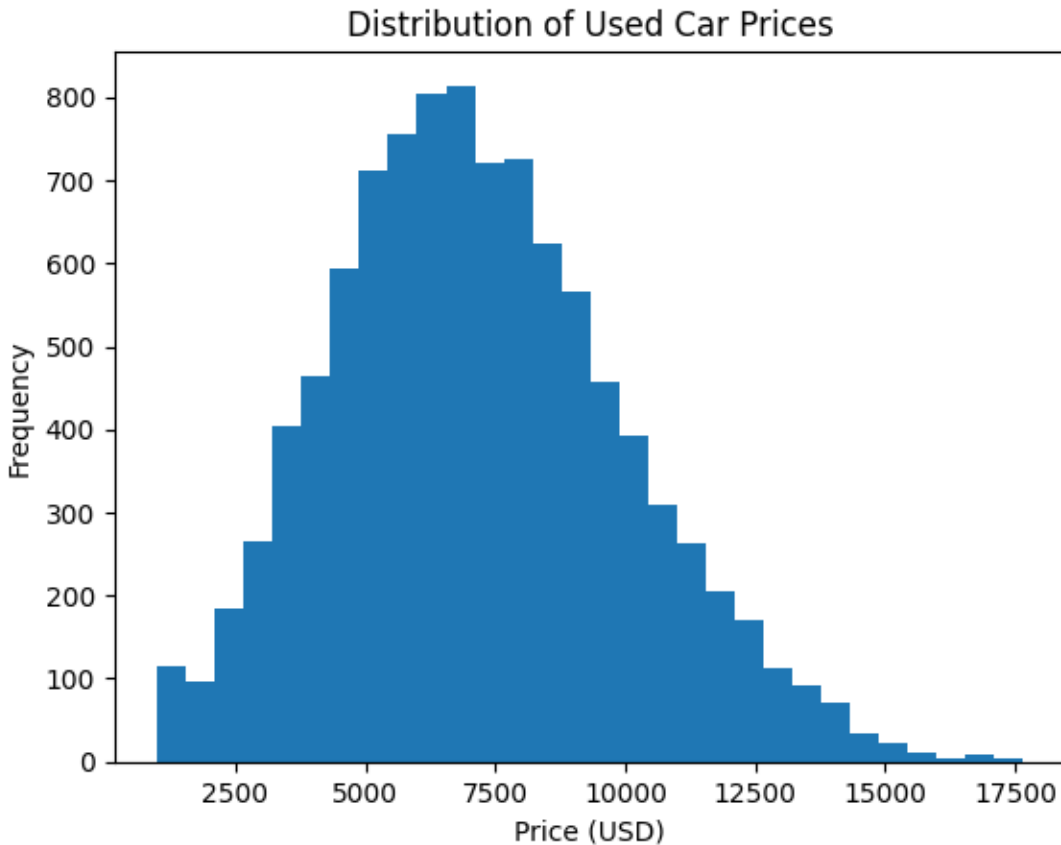
```
[89]:
```

	make_year	mileage_kmpl	engine_cc	fuel_type	owner_count
0	2001	8.17	4000	Petrol	4
1	2014	17.59	1500	Petrol	4
2	2023	18.09	2500	Diesel	5
3	2009	11.28	800	Petrol	1
4	2005	12.23	1000	Petrol	2

4. Exploratory Data Analysis

Exploratory data analysis (EDA) was performed to understand patterns, relationships, and trends within the dataset.

Distribution of Used Car Prices



The histogram shows that the distribution of used car prices is unimodal and slightly right-skewed. Most vehicles fall within a mid-range price band, while fewer vehicles appear at the higher end of the price spectrum.

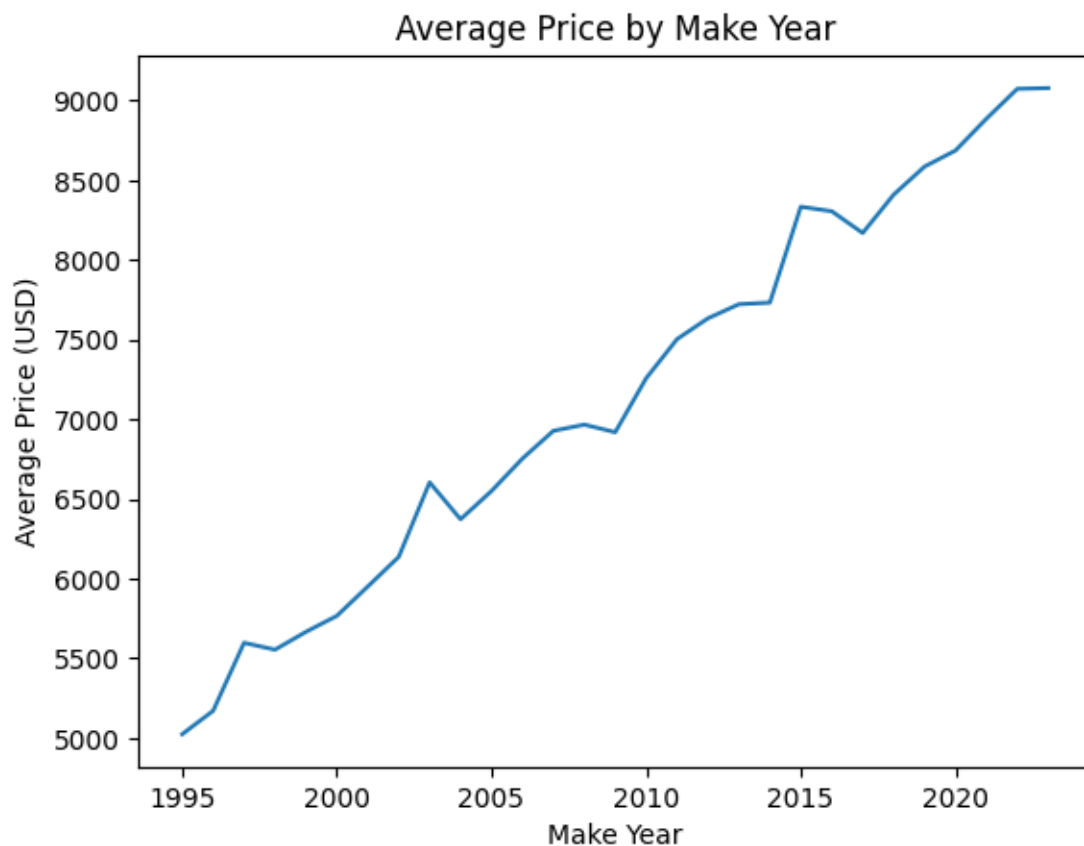
This suggests that the dataset is dominated by moderately priced vehicles, with a smaller number of high-value cars creating a longer tail on the right side of the distribution.

Mileage vs Price

The scatter plot shows a weak relationship between mileage and price. While there is some tendency for price to decrease as mileage increases, the relationship is not strong.

This indicates that mileage alone is not a strong predictor of price and must be considered alongside other features.

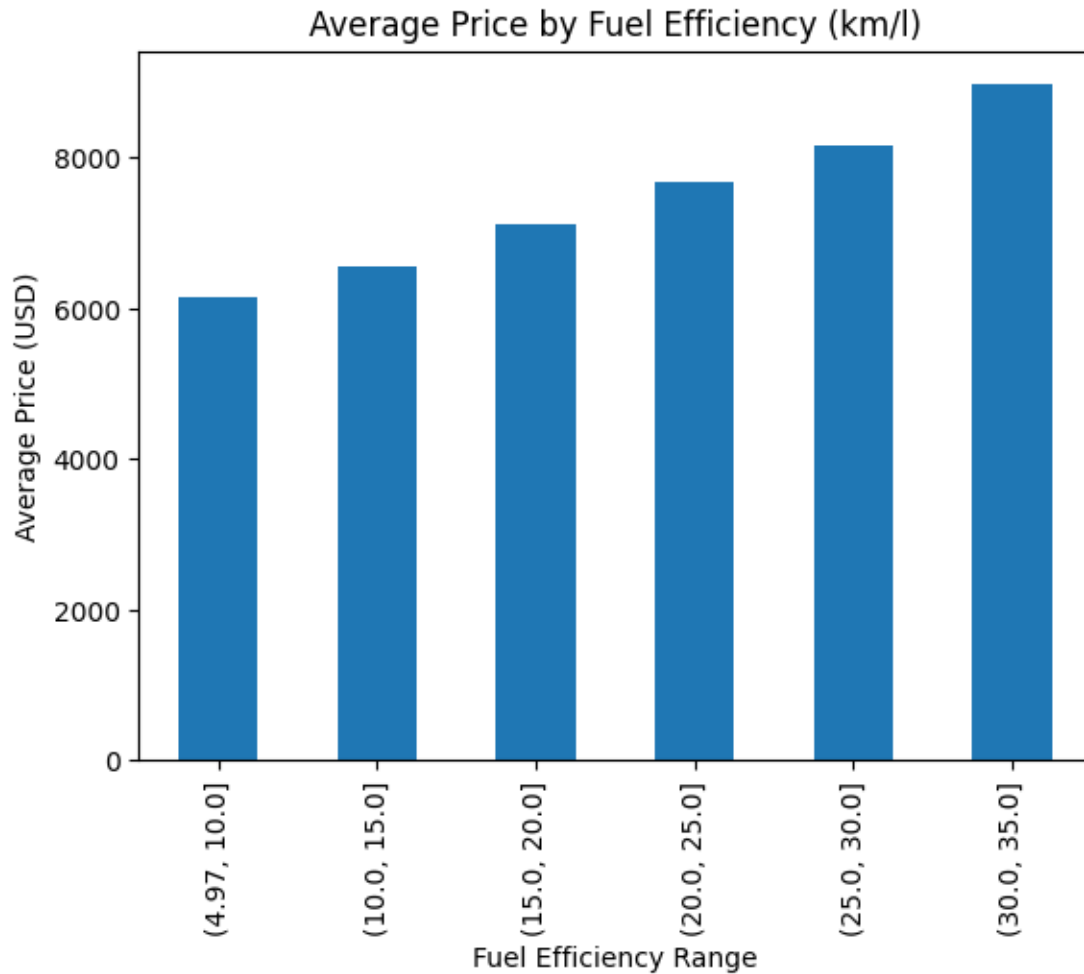
Average Price by Make Year



The line chart shows a clear upward trend in price as the make year increases. Newer vehicles tend to have higher average prices compared to older vehicles.

Although there are minor fluctuations, the overall trend strongly suggests that make year is an important factor in determining price.

Average Price by Fuel Efficiency

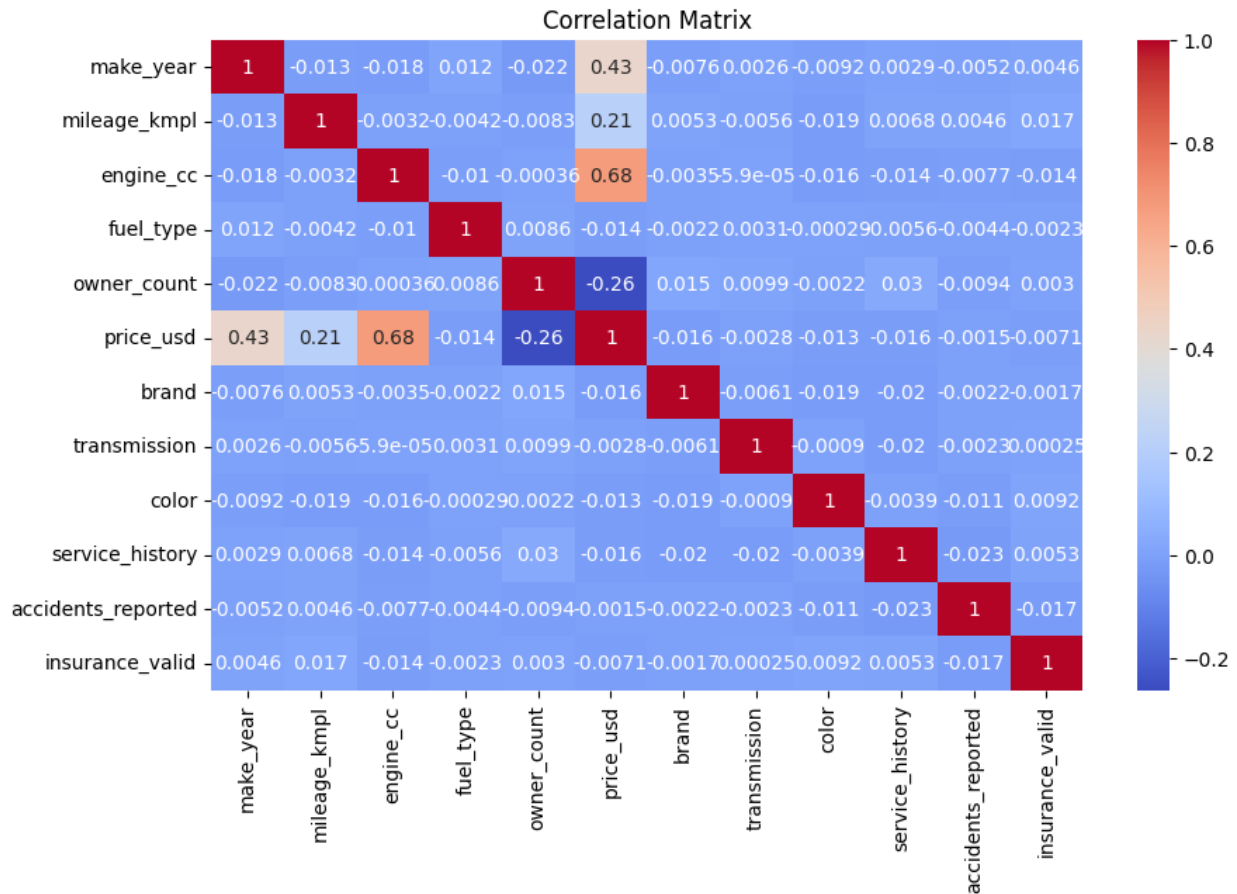


The bar

chart shows that average price tends to increase across higher fuel efficiency ranges. This indicates that vehicles with better fuel economy are associated with higher prices in this dataset.

However, this relationship may also be influenced by other factors such as make year or engine size, so it should not be interpreted as a direct cause-and-effect relationship.

Correlation Analysis



The correlation matrix shows the strength and direction of linear relationships between numerical variables.

Key observations include:

- Engine size has the strongest positive correlation with price
- Make year also has a positive correlation with price
- Owner count has a negative correlation with price

This means that larger engines and newer cars are generally associated with higher prices, while cars with more previous owners tend to have lower prices.

It is important to note that correlation only measures linear relationships and does not imply causation.

Data Modeling

Two machine learning models were used in this project:

Linear Regression

Linear Regression is a simple model that assumes a linear relationship between the input features and the target variable. It is easy to interpret but may not capture complex patterns in the data.

Random Forest Regressor

Random Forest is a more advanced model that uses multiple decision trees to capture non-linear relationships between features. It is more flexible and can model complex interactions between variables.

Both models were trained on the training dataset and evaluated using the testing dataset.

Evaluation and Results

The models were evaluated using three metrics:

- **Mean Absolute Error (MAE):** Measures the average difference between predicted and actual prices
- **Root Mean Squared Error (RMSE):** Penalizes larger prediction errors more heavily
- **R-squared (R^2):** Measures how well the model explains the variation in price

Linear Regression Results

- MAE: 1005.38
- RMSE: 1317.78
- R^2 : 0.782

Random Forest Results

- MAE: 959.83
- RMSE: 1220.94
- R^2 : 0.813

The Random Forest model performed better across all three metrics. It produced lower prediction errors and a higher R-squared value, indicating better overall performance.

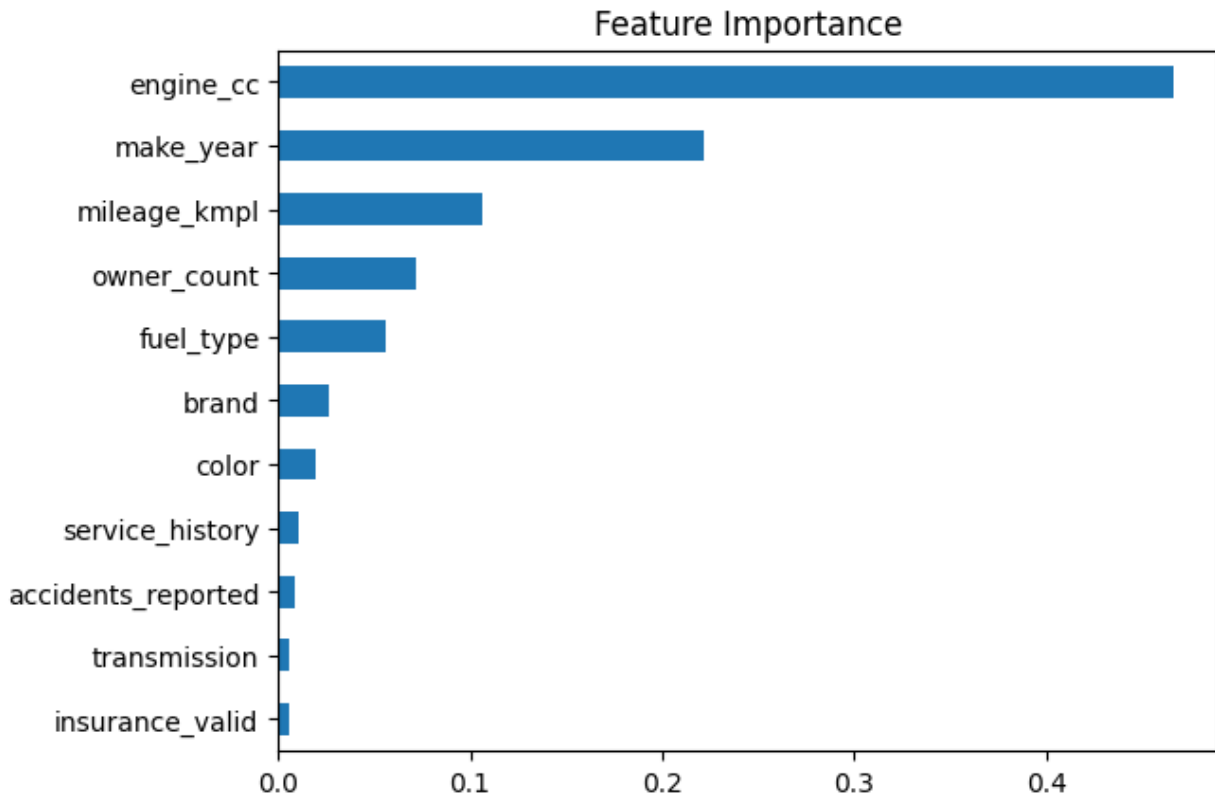
Results Interpretation

The results show that:

- Engine size is the most important factor influencing price
- Make year is strongly related to price
- Mileage has a smaller impact compared to other features
- Owner count negatively affects price

These findings are supported by both the correlation analysis and the feature importance results from the Random Forest model.

Feature Importance



The feature importance chart from the Random Forest model shows that engine size is the most influential feature, followed by make year and mileage.

This confirms that the model relies heavily on these variables when making predictions.

Model Demonstration

An example prediction was performed using the Random Forest model.

Sample input:

- Make year: 2015
- Mileage: 12.09 km/l
- Engine size: 5000 cc

- Fuel type: Diesel
- Owner count: 1

Predicted price: **\$12,354**

Actual price: **\$12,772**

The predicted value is very close to the actual value, demonstrating that the model performs well on unseen data.

```
[109]: # Example prediction
sample_car = X_test.iloc[0:1]
predicted_price = rf.predict(sample_car)

print("Sample Input:\n", sample_car)
print("Predicted Price:", predicted_price[0])
print("Actual Price:", y_test.iloc[0])
```

```
Sample Input:
      make_year  mileage_kmpl  engine_cc  fuel_type  owner_count  brand \
6252      2015          12.09      5000         0             1      6

      transmission  color  service_history  accidents_reported \
6252              0     5                0                   2

      insurance_valid
6252                0
Predicted Price: 12354.695400000002
Actual Price: 12772.12
```

Real-World Comparison to Industry Platforms

In the real world, used car pricing is handled by platforms such as Carvana, CarMax, and Kelley Blue Book (KBB). These platforms use data-driven approaches to estimate the value of vehicles based on a variety of factors.

Similar to this project, these systems rely on historical data and predictive models to determine pricing. Features such as vehicle age, mileage, engine specifications, and ownership history are commonly used to estimate a car's value. In this project, the machine learning model identified engine size, make year, and mileage as the most important factors, which aligns with how real-world pricing systems evaluate vehicles.

However, commercial platforms go beyond the scope of this project by incorporating additional data that was not available in this dataset. For example, companies like Carvana and CarMax may include:

- Vehicle condition (interior and exterior quality)
- Geographic location and regional demand
- Market trends and seasonal demand
- Detailed vehicle history reports

These additional features allow real-world systems to produce more accurate and personalized price estimates.

Another difference is that companies use more advanced and continuously updated models. While this project used Random Forest and Linear Regression, real-world platforms may use more complex machine learning techniques such as gradient boosting or deep learning, along with constant retraining as new data becomes available.

Despite these differences, the results of this project demonstrate a simplified version of how these systems operate. The ability of the Random Forest model to predict prices with reasonable accuracy shows that machine learning can effectively approximate real-world pricing strategies.

Overall, this comparison highlights that while the project model is simpler, it follows the same fundamental principles used by industry platforms: using data to identify patterns and make informed pricing predictions.

Conclusion

This project successfully applied machine learning techniques to predict used car prices. Both Linear Regression and Random Forest models were implemented and evaluated.

The Random Forest model performed better, indicating that more complex models are better suited for capturing relationships in this dataset.

The analysis showed that engine size, make year, and mileage are the most important factors influencing price.

Future improvements could include:

- Adding more features such as location or condition
- Tuning model parameters
- Testing additional models

Overall, this project demonstrates how data mining techniques can be used to solve real-world pricing problems.

References

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